Velocity graphs and components

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Read a velocity graph and describe motion in two directions.

SUCCESS CRITERIA - I can...

1. Use signed velocity-time area to find displacement.
2. Represent horizontal and vertical components separately.
3. Distinguish graph height, slope and vector magnitude.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “axis” to explain your model or result. Sentence starters are optional.

component: the part of a vector along one chosen axis

axis: a reference line used to measure a direction

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

Positions are 18, 12, 6 m at 0, 2, 4 s.
Predict position at 8 s.
Give the rate, an assumption and a future test.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

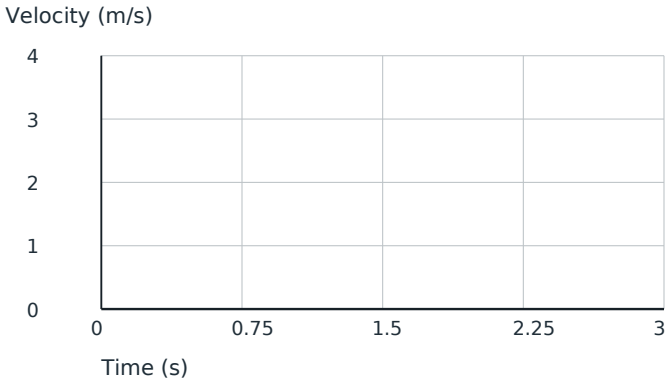
02 Worked example: explain the model

Reference: $v_x = v \cos\theta$; $v_y = v \sin\theta$.
θ measured from +x; degree mode. Signed velocity-time area gives displacement; total absolute area gives distance.

Worked reasoning: A cart travels at 3 m/s east for 2 s. The rectangle under its velocity-time graph has area 3×2=6 m, its displacement.

Your turn: Draw the cart's constant velocity on the axes. Shade the area between the line and the time axis.
Explain why multiplying the two axis units gives meters.

Sentence starter: Under the assumption …, the evidence … supports … because … .



Marzano check - circle: ① ② ③

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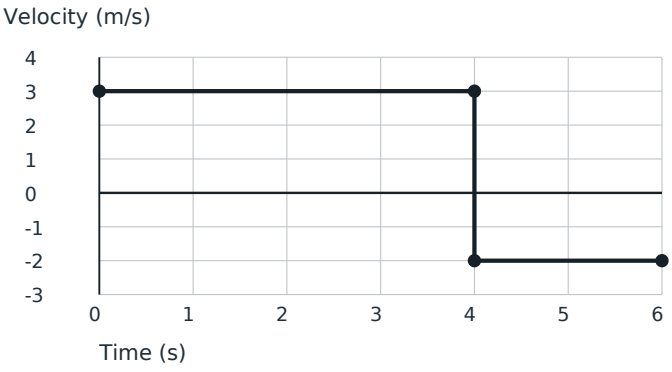
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Add signed areas

Use the graph.
Find displacement and distance from 0 to 6 s.
Describe the motion on each interval.

Sentence starter: I found ... [unit]. This value means ... in this situation.

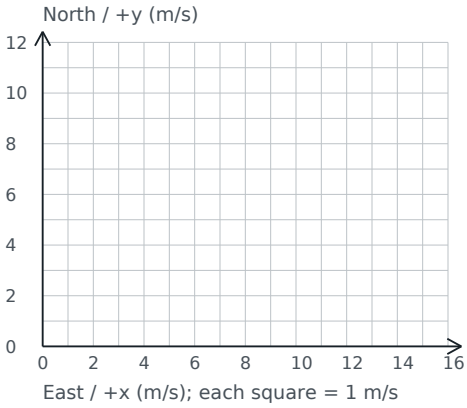


Marzano check - circle: 1 2 3

04 Resolve a northeast velocity

A drone flies at 14 m/s, 35° north of east.
Sketch its velocity and component triangle on the grid.
Predict which component is larger, then calculate both.

Sentence starter: Both ... . They differ in ... because ... .

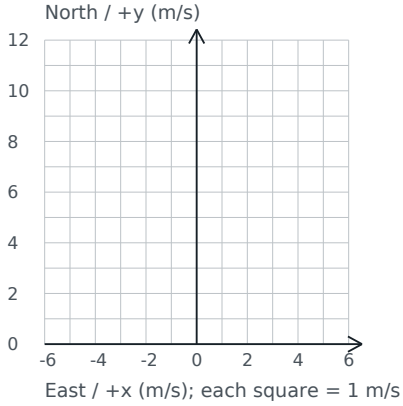


Marzano check - circle: 1 2 3

05 Change the reference axis

A boat travels at 10 m/s, 20° west of north. East is +x and north is +y.
Sketch the velocity, mark the angle from north, and find both signed components.
Explain your sine/cosine choices.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



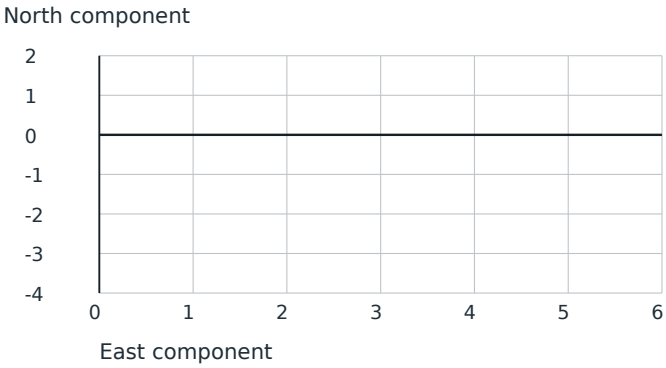
Marzano check - circle: 1 2 3

06 Draw velocity and displacement

An object has v=(+4,-3) m/s for 2 s.
Draw the velocity vector and calculate its speed, displacement vector, and displacement magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Two horizontal graphs

Explain the difference between a horizontal line at +5 on a position-time graph and a horizontal line at +5 on a velocity-time graph. Include units.

Sentence starter: Both ... . They differ in ... because ... .

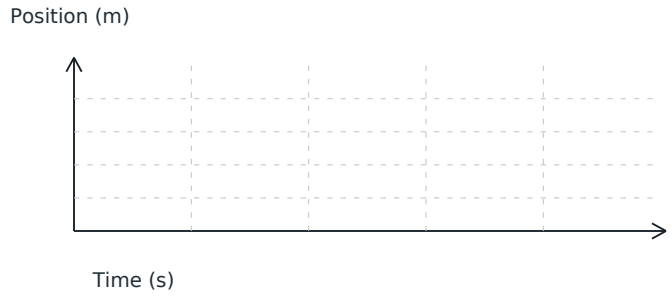
Marzano check - circle: 1 2 3

08 Review: connect an earlier idea

A straight position-time segment goes from (1 s, -3 m) to (5 s, 17 m).
Find velocity, interpret its units and predict x at 7 s if motion continues.
Label the model with the given positions and any prediction.

Word help: velocity = rate of position change, with direction

Sentence starter: I predict ... . My model supports this because ... .



Marzano check - circle: 1 2 3

09 Exit check

A velocity graph stays at +2 m/s for 3 s, then -1 m/s for 2 s.
Find displacement and distance.
State the information needed to interpret a vector angle of 30°.

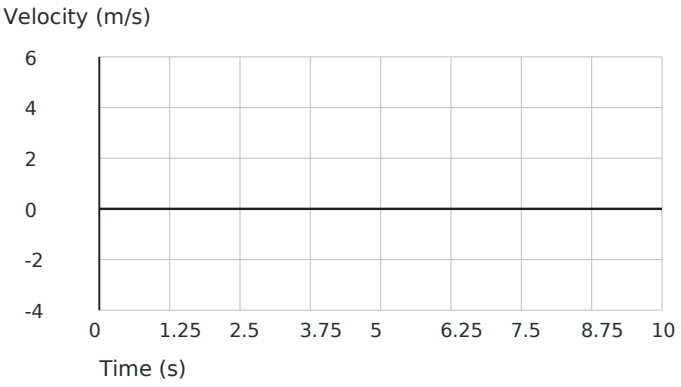
Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

10 STRETCH: test the model

A cart moves at +4 m/s for 3 s, then -2 m/s. How long must the second interval last to return to its start?
Sketch the graph and find average speed for the full trip.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3